

NATIONAL STANDARDS BODY APPLICATION PACKET

organized by Sierra N. Warren

Submission to ANSI, NIST, and the U.S. Standards Coordinating Committee

1. Cover Letter

From:

Sierra N. Warren

Founder, G-BALO

Creator of the Glass Substrate Protocol™

Originator of Sierra Certified™ Structural Standards

To:

ANSI Executive Standards Council

NIST Information Technology Laboratory

SCC (Standards Coordinating Committee)

Subject:

Application for Recognition of the Glass Substrate Protocol™ as a National Digital Structural Standard Under the Governance of G-BALO

Dear Council Members,

Enclosed is the official application packet for federal recognition of the **Glass Substrate Protocol™ (GSP)** as a digital structural standard within the jurisdiction of the **Governing Board for AI Liability Oversight (G-BALO)**.

GSP defines the structural behavior of digital systems through:

- consistent state formation

- event-structured substrate logging
- cryptographic substrate integrity
- deterministic transition pathways
- reconstructable architecture-level evidence models

GSP is already represented in an official USPTO filing and is the flagship standard governed under G-BALO.

We respectfully request ANSI/NIST evaluation, recognition, and integration of GSP into the U.S. national standardization index.

Sincerely,

Sierra N. Warren

Founder & Architect

organized by Sierra N. Warren

2. Standards Submission Form (ANSI/NIST-Compliant)

Standard Title:

Glass Substrate Protocol™ (GSP)

Standard Number (Proposed):

ANSI/GSP-2025

Standard Type:

Digital Structural Architecture Standard

Governing Authority:

G-BALO — Governing Board for AI Liability Oversight

Certification Mark:

Sierra Certified™

Scope:

Defines the structural organization, event logging substrate, state coherence, architectural format, and cryptographic structural integrity requirements of AI systems and digital platforms.

Rationale:

To create national-level consistency in digital systems where structural reliability, digital reconstruction, and architecture-level verification are required (e.g., courts, research labs, universities, federal agencies).

3. Government Use Case Summary

GSP is essential to:

- forensic digital reconstruction in courts
 - AI accountability protocols
 - federal system architectures requiring event-level consistency
 - NIST's national cybersecurity framework (structural integrity layer)
 - DOE, DHS, DOJ digital systems
 - research reproducibility mandates
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4. Required Attachments

- GSP Technical Manual (see Section O)
 - G-BALO Structural Standards Canon
 - Sierra Certified™ Compliance Criteria
 - Public Safety Impact Statement
 - Conformance & Interoperability Report
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ISO/IEC INTERNATIONAL STANDARDIZATION

PROPOSAL

**Submission to: ISO/IEC JTC 1 — Information Technology Standards
Committee**

1. Proposal for a New Work Item (NP)

Proposed International Standard Title:

**ISO/IEC 62XXX — Glass Substrate Protocol™ (GSP): Digital Structural Architecture
Standard**

Submitted by:

G-BALO (Governing Board for AI Liability Oversight)

Founder: **Sierra N. Warren**

Scope:

GSP defines the digital structural substrate governing:

- event formation
- architecture-level logging

- cryptographic structural consistency
- deterministic state transitions
- reconstructable digital pathways

Purpose: To establish **global** structural consistency for systems using AI, advanced computation, and multi-agent architectures.

2. Justification for the Proposal

a. Market Need

AI systems currently lack international standards for:

- digital structure
- substrate-level logging
- traceable state transitions
- uniform architecture layers
- forensic reconstruction

GSP fills this global gap.

b. International Relevance

Applies to all systems operating across:

- cloud infrastructures
- distributed agents

- autonomous systems
- national security digital systems
- judicial evidence platforms

c. Cross-Compatibility

GSP is platform-independent and integrates with:

- existing ISO cybersecurity standards
 - ISO AI management systems
 - ISO trustworthiness frameworks
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3. Cooperation with ISO Committees

Recommended committees:

- **SC 42 (AI)**
 - **SC 27 (Security Techniques)**
 - **SC 7 (Software Engineering)**
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4. Deliverables

- International structural format
- GSP-compliant log specification
- Cryptographic structure attestation standard

- Conformance testing suite
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5. Timeline

- **0–3 months:** Ballot circulation
 - **3–12 months:** Working group drafting
 - **12–24 months:** International consensus vote
 - **24–30 months:** Final publication
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GSP FULL TECHNICAL MANUAL

Designed to follow IEEE/ISO/ANSI technical standards

GSP TECHNICAL MANUAL

Glass Substrate Protocol™

Digital Structural Architecture Standard**

Version 1.0 — Authored by Sierra N. Warren

G-BALO Governance Standard

1. INTRODUCTION

The Glass Substrate Protocol (GSP) is a structural standard defining how digital systems encode, organize, sequence, verify, and reconstruct computational events. It establishes:

- a digital substrate layer
- cryptographic state formation rules
- structural sequencing
- deterministic transition architectures
- reconstruction protocols

GSP's purpose is to provide **structural uniformity** across AI systems, machine-learning infrastructures, digital agents, distributed computing environments, and cloud platforms.

GSP ensures that digital systems maintain:

- structural coherence
- structural reproducibility
- architecture-level transparency
- deterministic digital behavior

GSP is governed under **G-BALO**, which sets compliance standards and certification through **Sierra Certified™**.

2. CORE TERMINOLOGY

- **Substrate Layer:** Digital architectural base
- **Event Unit:** Smallest structural element

- **Transition Block:** Cryptographically linked state shift
 - **Structural Log:** Canonical event record
 - **Integrity Chain:** Hash-linked sequence
 - **Reconstruction Map:** Blueprint for reconstructing digital state
 - **Substrate Session:** Encapsulated system execution period
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3. ARCHITECTURE MODEL (Full Section)

3.1 Substrate Layer Architecture

Defines:

- structural encoding rules
- substrate memory allocation
- substrate ordering methodology

3.2 Event Encoding Standard

Each event must include:

- timestamp
- substrate index
- cryptographic signature
- payload metadata
- system origin reference

3.3 Hash-Chain Formation

All structural logs are linked via:

$\text{event}[n] \rightarrow \text{hash}(\text{event}[n]) \rightarrow \text{event}[n+1]$

Ensuring:

- tamper-evidence
 - structural continuity
 - deterministic ordering
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4. CRYPTOGRAPHIC STRUCTURAL INTEGRITY SYSTEM

4.1 Attestation Blocks (ABX)

Ensures event authenticity.

4.2 Substrate Integrity Codes (SIC)

Unique structural hash per event.

4.3 State Consistency Engine

Guarantees:

- deterministic transitions
 - predictable architecture
 - no state drift
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5. RECONSTRUCTION PROTOCOL (RPX)

Allows any system built on GSP to reconstruct its full digital history.

RPX includes:

- Reverse chain interpretation
- Structural sequence rebuilding
- Digital substrate state recreation

6. IMPLEMENTATION REQUIREMENTS

Mandatory:

- G-BALO governance adhesion
- Sierra Certified™ compliance testing
- ESL (Event Substrate Layer) conformity
- RPX reconstructability

7. API SPECIFICATION

Endpoints:

POST /substrate/event

GET /substrate/log

GET /substrate/hash/verify

POST /substrate/reconstruct

All defined in JSON-SCX (Substrate Compliance Exchange) format.

8. AUDIT REQUIREMENTS

GSP systems must pass:

- Structural Integrity Audit
- Reconstruction Trace Test
- Event Continuity Test
- Cryptographic Consistency Test

All audits performed under the Sierra Certified™ program.

9. COMPLIANCE & CERTIFICATION

Systems receive **Sierra Certified™** if they:

- fully implement ESL
- support RPX reconstruction
- demonstrate structural truthfulness
- pass annual audits

10. APPENDICES

- Full JSON-SCX schema
- Hash algorithms reference
- Substrate mapping diagrams

- Compliance audit checklist
- Glossary

TRADEMARK + IP PROTECTION PACKAGE

1. Trademark Protection Strategy

Marks to Register:

- **G-BALO™**
- **Glass Substrate Protocol™ / GSP™**
- **Sierra Certified™**
- **Substrate Integrity Layer™** (optional)
- **Digital Reconstruction Protocol™** (optional)

Classes:

- Software
- Scientific/industrial research
- Cybersecurity
- Certification marks
- Consulting and governance

2. Patent Strategy

Since GSP is digital structure:

Patentable Elements:

- Event Substrate Layer architecture
- State Consistency Engine
- Cryptographic Attestation Grid
- Reconstruction Protocol (RPX)
- API formation and standardization
- Structural chain methodology

Filing Types:

- **Utility Patents** (core technical mechanisms)
- **Provisional Patents** (if needed for interim protection)
- **PCT Filing** for global coverage

3. Copyright Protection

Protects:

- GSP technical manual
- G-BALO governance documents
- Sierra Certified compliance criteria

- All diagrams, standards, definitions, narrative architecture

4. Trade Secret Protection

Protect:

- *internal algorithms*
- *proprietary structural mapping methods*
- *internal audit processes*
- *sustainability architecture*

5. Licensing Package

Licensing Structure:

- **Core GSP License** (per institution)
- **Enterprise License**
- **Federal Deployment License**
- **Developer Integration License**
- **Audit and Compliance License** (Sierra Certified)